

THE UNIVERSITY OF HONG KONG
FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE

COMP7103 Data Mining

Date: May 20th 2021

Time: 6:30pm-8:30pm

There are 4 questions in total, each one giving either 30/100 or 20/100.

All questions are compulsory.

Candidates are permitted to use any online/electronic/printed/handwritten materials in the examination. Internet searching is allowed but crowdsourcing from group messages, online forums or social media, etc. is strictly forbidden.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Students can write/type/draw answer using computer and/or on paper. Students need to submit a single PDF file to OLEX."

Question 1 (k-means) (20 pts)

- a) (10 pts) Consider the following points in the Euclidean space: $P_1=(0,0)$, $P_2=(1,0)$, $P_3=(0,1)$, $P_4=(3,0)$, $P_5=(4,0)$, $P_6=(3,1)$, $P_7=(3,2)$, $P_8=(3,3)$, $P_9=(4,3)$. Run the k-means algorithm until convergence, starting with P_1, P_2 , and P_8 as initial centroids ($k=3$). Show all the steps, as well as the clusters obtained at the end of each step.
- b) (5 pts) Choose the initial centroids so that k-means terminates in one iteration.
- c) (5 pts) Does k-means always terminate for any possible choice of the initial centroids?

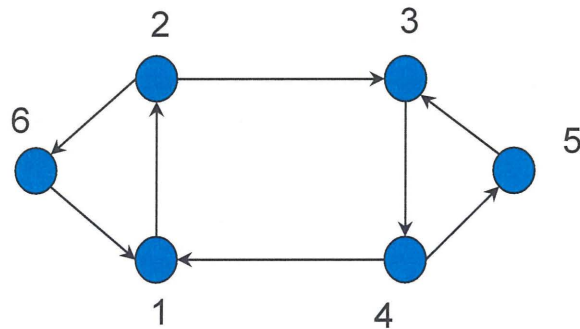
Question 2. (Decision Tree and Random Forest) (30 pts)

- a) (10 pts) Build a decision tree for the data reported in the table below. You should use the following rules: i) at each step compute the gini index for every possible split and select the split with the best gini value. (ii) stopping rule: when the gini value of a node is zero or no further split is possible; (iii) the class of a leaf node is determined by the majority rule with 0 being the default class. Show all the steps.
- b) (10 pts) After building the tree, prune the tree (if needed) so as to improve the generalization error= $\text{training error} + 0.5 * \text{number of leaves}$. Show all the steps. Explain the criteria used to stop the pruning.
- c) (5 pts) Predict the class of (A=1,B=1,C=1) using the tree before pruning. Predict the same data point using the tree obtained after pruning.
- a) (5 pts) We now build a trivial random forest containing only one tree. After bagging we obtain the 1st data point sampled twice, the 3rd data point and the 4th data point (N=4). Recall that at each split we sample k features without replacement. Suppose that in our case, k=2, while for the root we sample the features A and C. Build the corresponding tree using the same rules i),ii), iii) specified in a). You can break any tie arbitrarily. Show all the steps.

	A	B	C	Class
1	0	0	0	1
2	0	1	0	0
3	0	1	1	0
4	1	0	1	1
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0
8	1	1	1	1

Question 3. (PageRank) (20 pts)

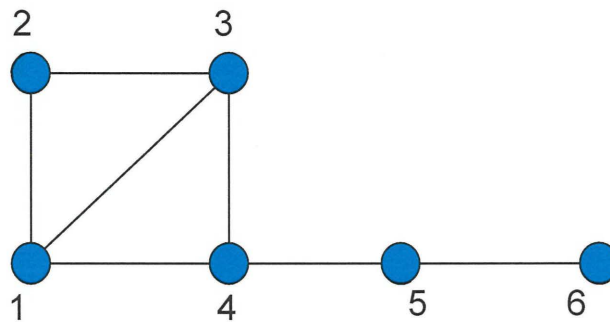
- a) (10 pts) Compute the PageRank vector for the graph below (without random jumps).
- b) (10 pts) Run the PageRank algorithm to compute the first two vectors starting from $(1/6, 1/6, 1/6, 1/6, 1/6, 1/6)$ (without random jumps).



Question 4. (Densest Subgraph) (30 pts)

Consider the graph G represented in the figure below.

- (7pts) Compute the k -core decomposition of such a graph.
- (7pts) Recall the greedy algorithm for computing a densest subgraph: at each step one node with minimum degree is removed; at the end we output the subgraph $H=(V_H, E_H)$ with maximum density $= |E_H|/|V_H|$ among all considered subgraph. Run the greedy algorithm on G and report the subgraph produced by such an algorithm.
- (7pts) What is the densest subgraph in G (i.e. it maximizes $|E_H|/|V_H|$)?
- (9pts) Provide a graph where the greedy algorithm finds a subgraph with density smaller than the density of the densest subgraph (approximately $1/2$ of the density of the densest subgraph to get full points.)



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